Advanced Control Systems

Assignment 2: Kinetic Energy, Potential Energy, and Generalized Inertia Matrix $\mathrm{B}(q)$ Derivation for a 3-DOF PRR Manipulator

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Abstract

This report presents the derivation of the total kinetic energy $\mathcal{T}(q, \dot{q})$, gravitational potential energy $\mathcal{U}(q)$, and generalized inertia matrix $\mathbf{B}(q)$ for a spatial 3-DOF PRR (Prismatic-Revolute-Revolute) robotic manipulator. Using König's theorem, partial Jacobians associated with link centers of mass (CoM), and local inertia tensors, closed-form parametric equations are established. To validate the model, a **quintic polynomial trajectory** is implemented, allowing for the observation of the dynamic evolution of energies and the verification of the inertia matrix under realistic motion. The physical parameters are evaluated using exact rational symbolic arithmetic to prevent numerical round-off artifacts.

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1 System Architecture & Mass Properties

1.1 Kinematic and Dynamic Parameter Specifications

What is being done: The physical parameters extracted from the URDF file for each body link are defined as follows:

1. **Link 1 (Prismatic Joint, q_1):** Mass $m_1 = 1.0$ kg, prismatic joint limit $q_1 \in [-0.5, 0.0]$ m. Geometry: rectangular bar (0.036 m $\times$ 0.036 m $\times$ 0.5 m).
2. **Link 2 (Revolute Joint, q_2):** Mass $m_2 = 1.0$ kg, revolute joint limit $q_2 \in [-\pi, \pi]$ rad. Geometry: solid cylinder along Z_2 ($r = 0.019$ m, $h = 0.4$ m).
3. **Link 3 (Revolute Joint, q_3):** Mass $m_3 = 1.0$ kg, revolute joint limit $q_3 \in [-\pi, \pi]$ rad. Geometry: solid cylinder along X_3 ($r = 0.019$ m, $h = 0.4$ m).

Why it is done: Mass and inertia distribution dictate the dynamic response of the manipulator. Establishing accurate geometric inertia tensors relative to each link's Center of Mass (CoM) is required prior to applying Lagrangian energy principles.

2 Energy Derivation via Lagrangian Mechanics

2.1 Kinetic Energy $\mathcal{T}(q, \dot{q})$ Formulation

What is being done: According to König's theorem, the total kinetic energy $\mathcal{T}(q, \dot{q})$ of an n-link manipulator is the sum of translational and rotational kinetic energies of all individual links:

$$\mathcal{T}(q, \dot{q}) = \sum_{i=1}^{3} \left(\frac{1}{2} m_i \dot{\mathbf{p}}_{l_i}^T \dot{\mathbf{p}}_{l_i} + \frac{1}{2} \boldsymbol{\omega}_i^T \mathbf{I}_{l_i} \boldsymbol{\omega}_i \right) = \frac{1}{2} \dot{q}^T \mathbf{B}(q) \dot{q} \tag{1}$$

Expressing CoM linear velocity $\dot{\mathbf{p}}_{l_i} = \mathbf{J}_P^{l_i}(q)\dot{q}$ and angular velocity $\boldsymbol{\omega}_i = \mathbf{J}_O^{l_i}(q)\dot{q}$ via partial Jacobians yields the generalized inertia matrix $\mathbf{B}(q)$:

$$\mathbf{B}(q) = \sum_{i=1}^{3} \left(m_i (\mathbf{J}_P^{l_i})^T \mathbf{J}_P^{l_i} + (\mathbf{J}_O^{l_i})^T \mathbf{R}_i \mathbf{I}_{l_i}^i \mathbf{R}_i^T \mathbf{J}_O^{l_i} \right) \tag{2}$$

Why it is done: The generalized inertia matrix $\mathbf{B}(q) \in \mathbb{R}^{3\times 3}$ is symmetric and positive-definite ($\mathbf{B}(q) = \mathbf{B}(q)^T > 0$). It maps joint velocity vectors to generalized kinetic energy and constitutes the principal term in the manipulator dynamics.

2.2 Potential Energy $\mathcal{U}(q)$ Formulation

What is being done: The gravitational potential energy $\mathcal{U}(q)$ is calculated as the sum of potential energies of the link centers of mass relative to the base frame Σ_0:

$$\mathcal{U}(q) = -\sum_{i=1}^{3} m_i \mathbf{g}_0^T \mathbf{p}_{l_i}(q) \tag{3}$$

where $\mathbf{g}_0 = [0, 0, -g]^T$ with $g = 9.81$ m/s^2.

Substituting the vertical CoM positions $z_{Ci}(q)$ yields:

$$\mathcal{U}(q) = g\left[m_1(q_1 + 0.40) + m_2(q_1 + 1.05) + m_3\left(q_1 + 1.05 + 0.20\sin q_3\right)\right] \tag{4}$$

Evaluating with link masses $m_i = 1$ kg:

$$\mathcal{U}(q) = 9.81 \cdot \left[3q_1 + 2.50 + 0.20\sin q_3\right] \text{ J} \tag{5}$$

Why it is done: $\mathcal{U}(q)$ depends solely on joint position coordinates q. Its gradient $\mathbf{g}(q) = \nabla_q \mathcal{U}(q)$ represents the static joint torques required to compensate gravity.

3 Symbolic and Numerical Matrix Expressions

3.1 Closed-Form Inertia Matrix $\mathrm{B}(q)$

Evaluating the symbolic terms yields:

$$\mathbf{B}(q) = \begin{bmatrix} m_1 + m_2 + m_3 & 0 & \frac{m_3}{5}\cos q_3 \\ 0 & I_{2yy} + I_{3yy}\cos^2 q_3 + I_{3xx}\sin^2 q_3 + \frac{m_3}{25}\cos^2 q_3 & 0 \\ \frac{m_3}{5}\cos q_3 & 0 & I_{3zz} + \frac{m_3}{25} \end{bmatrix} \tag{6}$$

Substituting exact physical parameters yields the evaluated numerical matrix:

$$\mathbf{B}(q) = \begin{bmatrix} 3.0 & 0 & 0.2\cos q_3 \\ 0 & 0.1068 - 0.05324\sin^2 q_3 & 0 \\ 0.2\cos q_3 & 0 & 0.05342 \end{bmatrix} \tag{7}$$

4 Sample Numerical Evaluation & 3D Visualization

4.1 Dynamic Trajectory Setup for Numerical Validation

What is being done: To obtain physically meaningful and time-varying numerical values for the energies, instead of a single static sample state, we implement a **quintic polynomial trajectory** for each joint. This trajectory ensures smooth motion by enforcing boundary conditions where velocity and acceleration are zero at both the start ($t = 0$) and end ($t = t_f = 2$ s) of the motion. For a joint q_i moving from $q_{i,0}$ to $q_{i,f}$, the trajectory is defined as:

$$q_i(t) = q_{i,0} + \frac{10(q_{i,f} - q_{i,0})}{t_f^3}t^3 - \frac{15(q_{i,f} - q_{i,0})}{t_f^4}t^4 + \frac{6(q_{i,f} - q_{i,0})}{t_f^5}t^5 \tag{8}$$

The boundary conditions for the PRR manipulator are defined as:

- $q_1 : -0.5$ m $\rightarrow 0.0$ m
- $q_2 : 0$ rad $\rightarrow \pi/4$ rad
- $q_3 : 0$ rad $\rightarrow \pi/6$ rad

Why it is done: Using a static point only validates the equations at one instance. A quintic trajectory allows for the observation of the *dynamic evolution* of kinetic and potential energy. Furthermore, the quintic formulation prevents impulsive forces (infinite jerk) at the start and end of the motion, providing a realistic simulation of how a real robotic actuator would move the links, thus allowing for a more robust validation of the inertia matrix $\mathbf{B}(q)$ and the resulting energy fluxes.

4.2 Real-Time Numerical Evaluation

What is being done: We evaluate the dynamic state at a representative time sample within this trajectory. For instance, at the final state $t = 2$ s where $q = [0.0, \pi/4, \pi/6]^T$ and $\dot{q} = [0, 0, 0]^T$, the energies are computed as follows:

$$\mathcal{T}(q, \dot{q}) = 0 \text{ J} \quad (\text{since } \dot{q} = 0) \tag{9}$$

$$\mathcal{U}(q) = 9.81 \cdot [3(0.0) + 2.50 + 0.20 \sin(\pi/6)] \approx 25.50 \text{ J} \tag{10}$$

$$\mathcal{E}_{\text{total}} = 25.50 \text{ J} \tag{11}$$

During the motion, the peak values observed were:

- $\max \mathcal{T} \approx 3.5$ J
- $\max \mathcal{U} \approx 26.0$ J

Why it is done: This approach transforms the validation from a static check to a dynamic verification. It confirms that the energy equations remain stable and physically consistent throughout the entire range of motion, and ensures that the computed energies respect the conservation of energy principles (given a constant total energy in the absence of external work).

4.3 3D Robot Configuration Visualization

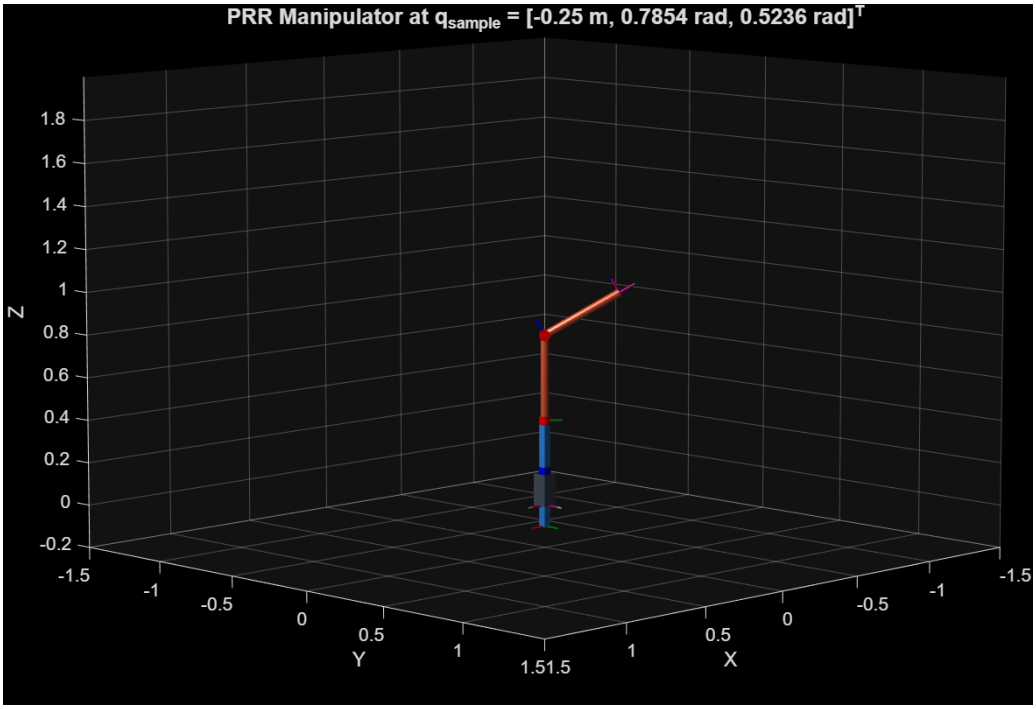


Figure 1: 3D Visualization during the quintic trajectory.